

Validating AI-Driven Nudge Recommendations: A/B Testing Two-Tower and Bandit Models in Simulated Digital Banking Environment

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ABSTRACT While radical moves towards personalization characterize increasingly ambitious digital banking campaigns, recommender systems are typically based on either collaborative filtering or content-based filtering, and they often prove ineffective due to a lack of behavioral awareness, as well as issues such as high sparsity, cold start, and historical signal sparsity. To overcome this gap, this work proposes an AI-based Nudge Recommendation Model that incorporates a Two-Tower Network (TWN) for static personalization along with a Multi-Armed Bandit (MAB) for real-time adaptive nudge optimization. We tested the model using a controlled A/B experiment in a mobile banking simulator with verified bank customers. Results show substantial behavioral improvements in the treatment group using the AI-driven system, including an increase in overall purchase behavior (48.6% to 62.07%) and a more than fourfold rise in recommendation-following rates (13.6% to 52.87%). Statistical testing confirms the significance of these differences ($\chi^2 = 6.49$, $p < 0.0108$). These findings validate the causal impact of personalized and adaptive nudging on user decision behavior and provide a reproducible machine learning behavioral economics model with implications for improving digital banking engagement.

INDEX TERMS A/B testing evaluation, AI in digital banking, multi-armed bandit (MAB), nudge recommendation model, two-tower network (TWN).

I. INTRODUCTION

The banking industry is progressively using artificial intelligence to improve customer engagement and campaign results in the digital age [1]. Nudging technique is one of the promising approaches for the application of behavioral economics that has been widely implemented in offline environments [2], [3]. Nudging refers to subtle modifications in choice architecture that steer users toward better decisions without restricting their freedom of choice [4]. In digital banking, nudging is particularly relevant due to high perceived risk, low product familiarity, and decision fatigue commonly observed among retail banking customers [5]. However, to optimize the timing and relevance of these nudges, AI models must be intelligent and responsive [6]. Static recommendation systems have been the focus of previous research, the drawback being that they

have not been able to fully account for changes in user preferences and behavior over time [7].

The integration of behavioral nudge theory with static and adaptive recommendation models is still not fully developed, and this research suggests a combined model that uses a Multi-Armed Bandit algorithm for fast adjustment and a Two-Tower Network for fixed personalization to fill the gap [8], [9], [10]. The purpose of this research is to evaluate the previously developed AI-based nudge recommendation model. Through A/B testing in a simulated mobile banking application designed to mirror real banking features, allowing users to explore products and perform purchase-like actions, this research aims to provide empirical evidence regarding the model's performance in delivering relevant and timely nudges in digital banking campaigns [11].

While simultaneously improving personalization and adaptability, the integrated model proposes a strategic approach to maintain a trade-off between exploration and exploitation, a critical aspect of financial decision space [12]. Addressing cold-start problems and sparsity of user behavior, the Two-Tower Network ensures personalization at the baseline level through user and product profiles that capture historical information as well [13]. On the other hand, the Multi-Armed Bandit algorithm continually integrates real-time feedback from users, so the recommendations can change depending on the history of user actions over time and the changing contextual environments [14].

II. LITERATURE REVIEW

A. STATIC AND ADAPTIVE MODELS OF AI-DRIVEN RECOMMENDATION SYSTEMS

Recent developments in AI-based recommendation systems have made profound impacts on digital banking [7], [8]. The advancement of AI has enabled large-scale adoption of personalization through recommender engines [6]. Most traditional systems are either collaborative filtering or content-based [7], [15]. Two-Tower Model (TWN) architecture is a solution to learn deep discrete representations of users and items in a very powerful way [15], [16]. Typically, these embeddings are combined via a dot product to compute some relevance score, allowing for scalable recommendations as user and item representations are independent [17], [18]. TWNs enable efficient static personalization based on users' interests and products' characteristics while avoiding sparsity in users' behaviors [15], [19]. As the static recommendation backbone of the hybrid model, TWNs are used to obtain contextually relevant product recommendations for every user [15].

Although TWNs can learn a good ordering of items, it cannot dynamically adapt. The Multi-Armed Bandit (MAB) algorithms fill this gap by balancing exploration and exploitation [20], [21], leveraging user interactions for online learning and dynamic calibration of recommendations over time [22]. Using this hybrid architecture, digital banking applications increasingly use behavioral economic principles based on nudge theory to layer a further degree of personalization on top of a recommendation system, using cognitive and emotional nudges such as social proof, defaults, loss aversion, and framing effects [2], [3], [23].

B. NUDGE THEORY AND BEHAVIORAL ECONOMICS

Behavioral economics studies the effects of psychological, cognitive, emotional, cultural and social factors on the decisions of individuals and institutions and how those decisions vary from those implied by classical economic theory [24]. The discipline is based on the concept of nudging, as conceived by Thaler and Sunstein [25]. It is all about indirect changes in decision architecture that help people choose while maintaining freedom of choice [26]. Nudges are increasingly applied in digital banking to encourage desirable

behaviour such as responsible spending, saving and product engagement, by making changes to the way in which options are framed or presented [2]. We implement nine established nudging dimensions: default options, personalization, framing effect, social proof, loss aversion, gamification/incentives, just-in-time nudging, and saliency, as well defined by previous literature [7]. Despite nudging has been studied in intersection with behavioral economics, nudging implemented in the context of AI-based recommendation systems is still limited [27]. By integrating these nudge principles into the AI system in ongoing personal real-time interaction with the user with respect to the user's behaviour and feedback, this gap is addressed. As a result, the system can provide real-time, context-aware considerations for each user [28].

Classical financial consultation methodologies generally apply universal rules, without considering complex psychological and behavioral profiles of end users [29]. By integrating nudge design into the AI recommendation architecture, the system is now able to personalize what to recommend and when and how to present a recommendation. This leads to increased relevance and responsiveness especially when it comes to high-stake financial decisions where users usually suffer from cognitive load and perception of risk [30].

C. LIMITATIONS OF EXISTING APPROACHES AND RESEARCH GAP

Historically, recommended systems in financial services have become global unique identification numbers (GUID) on either collaborative filtering (CF) or content-based filtering (CBF). These approaches have been shown to work well in dense interaction history settings, such as e-commerce or entertainment platforms, but they fundamentally limit their applicability to the banking context [7], [15]. First of all, the cold-start problem is common among bank users since most bank customers perform financial product transactions rarely, resulting in sparse behavioral signals that CF models need to calculate similarity [8], [16]. First of all, the cold-start problem is common among bank users since most bank customers perform financial product transactions rarely, resulting in sparse behavioral signals that CF models need to calculate similarity [15], [19]. Third, and most critical for this study, CF and CBF do not incorporate psychological or behavioral mechanisms, meaning they cannot model how nudges, such as framing, personalization cues, loss aversion, or salience, shape financial decisions [2], [3], [23]. These constraints reduce their usefulness in environments where perceived risk, trust, and cognitive load strongly influence user actions [8].

Beyond classical recommenders, existing AI-based personalization approaches in digital banking also show important gaps. While machine learning has been widely applied to credit scoring, fraud detection, and targeted marketing, very few studies explicitly integrate behavioral economics into algorithmic recommendation workflows [6], [31]. Current models optimize engagement metrics, but rarely account for

the distinct behavioral effects of different nudge mechanisms [8], [32]. Likewise, adaptive learning methods such as Multi-Armed Bandits have been employed in marketing optimization, yet seldom in settings where each action represents a behavioral nudge type designed to influence financial choices [33].

A further limitation in the literature is the lack of empirical validation with verified bank customers. Previous studies often rely on simulated settings, hypothetical vignettes, or records from a historical log rather than observing actual decision-making among individuals who are real customers of a financial institution [34]. Although there has been progress in recommender systems, adaptive learning, and digital nudging, to the best of our knowledge there are no empirically validated nudge-aware AI recommendation model in banking that integrate these three components, specifically: (1) static personalization, (2) real-time adaptive learning, and (3) explicit behavioral design tested under a controlled experimental setting with verified bank customers [8]. Previous studies have mainly benchmarked collaborative filtering, content-based filtering with behavioral design, and contextual bandits without behavioral design. None of these approaches jointly integrate static personalization with adaptive nudge optimization. This study thus offers an empirical hybrid baseline never tested previously in banking contexts.

III. RESEARCH METHODS

A. DESIGN SCIENCE RESEARCH METHODOLOGY (DSR)

The research is primarily based on Design Science Research (DSR) [35] (Fig. 1), which emphasizes the construction, validation, and evaluation of a new artifact (in this case, an AI Nudge recommendation model for digital banking campaigns). This study aligns with primary DSR phases starting from identifying the problem and defining the objectives to the design and development of the model based on six building blocks obtained from literature and expert opinions. The overall experimental design and evaluation flow, including the control and treatment groups, model development, and outcome measurement, are illustrated in Fig. 2. Although the general research framework fits well with the DSR iterative build-and-evaluate cycle, in this article we focus especially on the evaluation phase by performing A/B testing. The A/B test is the main tool to empirically validate if the model leads users to make better decisions. Primarily, this article addresses Research Question 3, concerning the performance of the AI-based Nudge Recommendation Model that we apply in a digital banking environment. The earlier stages, represented by RQ1 and RQ2, centered on defining the model’s key components and developing the system, including a mobile banking prototype designed to mirror real application behavior. Here, the intention is to move beyond design and theory and instead highlight how well the model works in influencing user decisions. By focusing on A/B testing results, user interactions, and performance outcomes, this article shows the model’s real impact in a practical, measurable way.

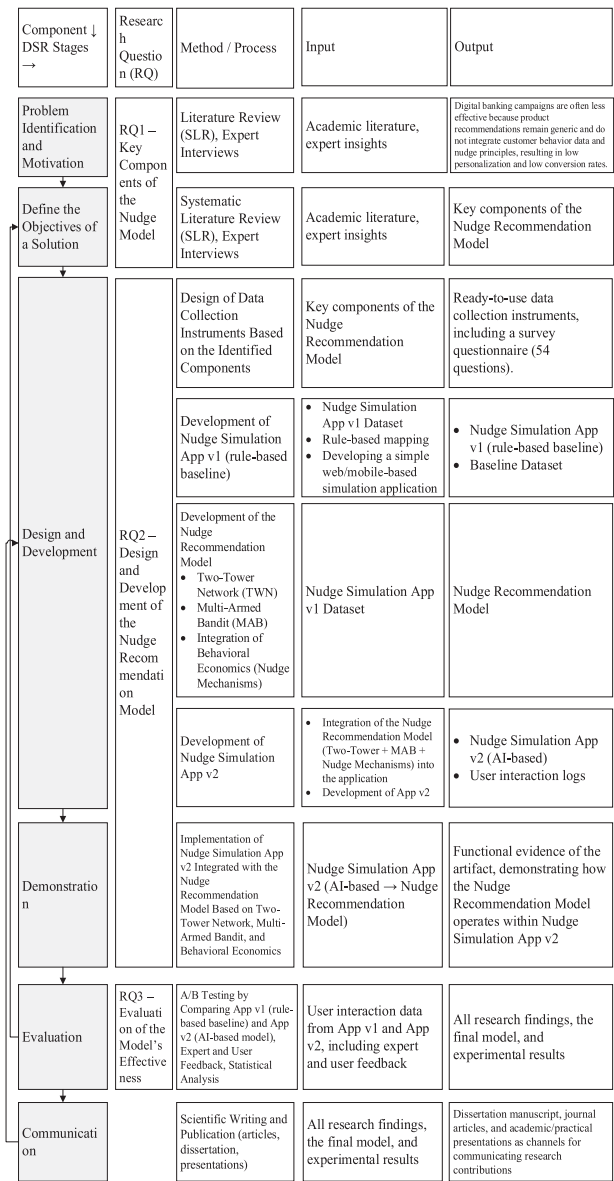


FIGURE 1. Design science research methodology (DSRM).

B. A/B TESTING PROCEDURE

A controlled A/B testing procedure was implemented to evaluate the effectiveness of the AI-driven nudge model, consistent with best practices in experimental design [3]. Two application environments were deployed: App v1 (Control), which used rule-based nudge assignment, and App v2 (Treatment), which incorporated the Two-Tower Network and Multi-Armed Bandit for personalized and adaptive nudging. Both environments followed the same simulation flow, product choices, page structure, and decision points, ensuring that the only manipulated variable was the nudge-selection mechanism [8].

Participants were segmented based on the version of the application they interacted with. Group A consisted of 214 verified bank customers who used App v1 during the initial

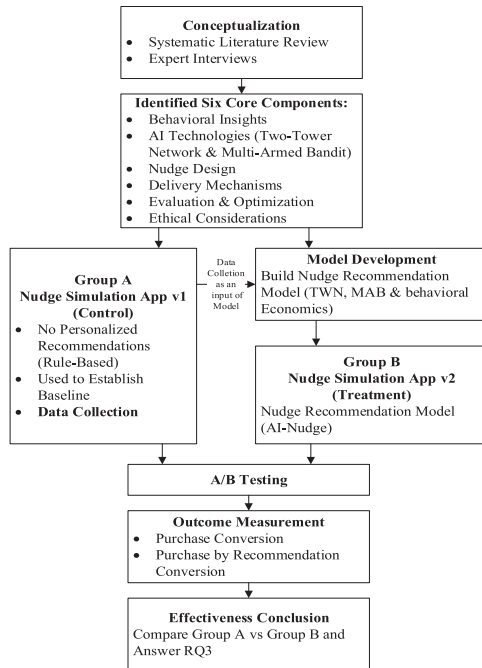


FIGURE 2. A/B testing research flow diagram.

testing window. Group B consisted of 174 verified customers who used App v2 during the second testing window. No participant appeared in both groups, ensuring independence and preventing cross-conditioning contamination [36], [37]. The system recorded important variables for each recommendation event, including the recommended item, assigned nudge type, user choices, timestamp, and purchase results. These logs allowed us to compare across the two experimental conditions directly for our main purchase outcomes and recommendations, which were the main metrics used in the evaluation of the treatment [38].

The A/B testing in this study used a sequential quasi-experimental design, which reflects the two-stage development and deployment cycle of the Nudge Simulation App. App v1 (rule-based nudging) was released during the first data-collection phase. These were subsequently used to train and calibrate the Two-Tower Network and Multi-Armed Bandit models. Once the AI model was built, a second version of the app, App v2, was implemented to record data from the treatment group within a subsequent testing period. Participants were therefore separated into two temporally distinct cohorts, as the two application versions were released at different times. To ensure all experimental conditions were independent, App v1 users were removed from the users available to include in the App v2 dataset. This ensured that no participant was a member of both groups. By separating users into Group A and Group B, these represent mutually exclusive user populations that were exposed to two different versions of the simulation environment, in accordance with best practices in quasi-experimental evaluation of sequential deployments of a system. Our treatment sample (Group B)

was smaller than the control sample, but the architecture was trained exclusively on the larger baseline dataset (Group A), and Group B was used only for evaluation. Such separation prevents information leakage and adheres to sequential A/B deployment conventions. This design ensures that learning effects are not transferred across cohorts while preserving operational realism in staged system deployment.

Due to the lack of overlap in time periods from which control and treatment samples were drawn, this evaluation is better considered a sequential quasi-experiment than a parallel random A/B test. Therefore, the research design does not allow us to make strong claims about causality, and differences measured should be interpreted in light of time period effects. However, the scale and consistency of differences in behavior indicate that time cannot be the sole driver of this effect.

In addition to the rule-based baseline considered in Group A and the AI-based system evaluated in Group B, we introduce a TWN-only ablation baseline. In this regime, the recommendations are made by the Two-Tower Network itself with no multi-armed bandit adaptation or behavioral nudging. This ablation is done to extract the effect of behavioral nudging and adaptive bandit layer beyond static personalization. It thus constitutes an intermediate benchmark between static personalization and fully adaptive, behavior-optimized recommendation.

C. DATA COLLECTION AND SIMULATION ENVIRONMENT

Data for this study was collected through a structured simulation environment designed to capture user preferences, decision patterns, and behavioral responses during the financial product selection journey. All participants answered a 54-item questionnaire related to demographic characteristics, financial behavior, risk perception, spending orientation, saving patterns, and digital engagement behavior. This type of multi-dimensional survey instrument is typically used to build psychographic and behavioral profiles that feed into personalized modeling [39]. This set of responses made up for the main features used while building the user embedding within the personalization model. Participants then went through a simulation in either App v1 (Control) or App v2 (Treatment), depending on their group assignment after the survey. Using a common Supabase data pipeline, the app automatically recorded all events throughout the interaction, including recommended items, assigned nudge types, timestamps, and purchase outcomes. These logs provided the behavioral ground truth used for evaluating baseline decision patterns in Group A and for training and validating the adaptive nudge model deployed in Group B.

D. NUDGE MECHANISMS

From the systematic literature review conducted in this research developed in accordance with the systematic review of methods [15], which synthesized 100 empirical studies on digital nudging, behavioral economics, and financial decision support systems, the wider literature highlights numerous

nudging strategies that could impact user choice. Framing nudges influence decision making by framing information in terms of features of a choice, and personalization cues adjust messages based on characteristics of user to make it relevant. By framing a loss instead of a gain, loss aversion compels people to act, while incentives give people either money or non-monetary rewards to push them towards desirable behaviours. Social proof uses information on how peers behave to inform individual decisions, and saliency enhancement increases visibility of key information to users. Gamification brings similar activities from gaming to boost engagement, default options simplify decision-making by pre-selecting an optimal option, and just-in-time nudges are interventions delivered now of decision-making.

Nevertheless, this article targets nine mechanisms that have been consistently reported across digital interventions and that have shown large effect sizes on behavior in the financial domain: personalization, default option, framing, social proof, loss aversion, incentives, gamification, just-in-time nudging, and saliency [3], [7], [40]. These nine mechanisms were selected because they appear with high frequency in the SLR, are directly relevant to savings, investing, insurance, and risk-related decisions, and can be operationalized as message-level cues suitable for integration into the model. The mapping of nudges to financial products follows a structured coding derived from the SLR taxonomy [15], in which each product may be associated with multiple nudges depending on its attributes, behavioral drivers, and decision patterns described in prior literature. Savings products, for instance, are often consistent with default options and loss aversion, investment products with framing and social proof, and insurance products with loss aversion and incentives [40]. These mapped nudges subsequently define the action space of the Multi-Armed Bandit, while the Two-Tower Network provides the relevance scores used to personalize which nudge mechanism is most appropriate for each user at each moment.

E. MODEL ARCHITECTURE AND TRAINING

The Nudge Recommendation Model has two main components: a Two-Tower Network (TWN) to capture static personalization and a Multi-Armed Bandit (MAB) to dynamically select and adapt nudges to users in real-time. The TWN consists of a User Tower, which transforms demographic attributes, behavioral indicators, and questionnaire responses into a latent user embedding, and a Product Tower, which encodes product attributes into a corresponding item embedding. Both embeddings are projected into a shared vector space, and a dot-product similarity generates a personalized relevance score for each user-product pair, following standard dual-encoder practices in large-scale recommendation systems [15], [16]. This score is not shown to the user directly; instead, it functions as a prior signal that informs the bandit's decision-making process [8].

TWN was trained in preference labels derived from baseline interactions in Group A, allowing the model to learn stable representation of user preference prior to treatment

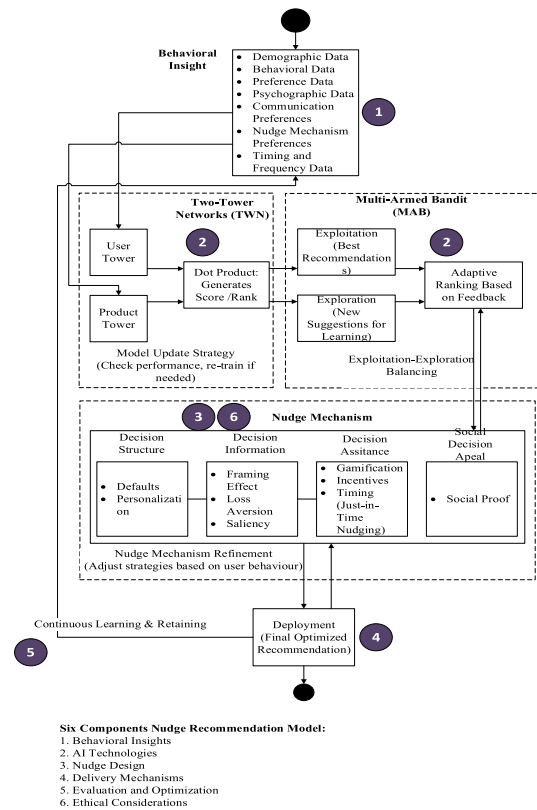


FIGURE 3. Integrated architecture of the nudge recommendation model (TWN + MAB).

environment deployment. We set binary cross-entropy as a loss function and Adam as an optimizer and used early stopping to avoid overfitting. Because this study focuses on decision behavior rather than deep model optimization, architectural details are intentionally kept concise while ensuring reproducibility.

The relevance score generated by the TWN is passed to the Multi-Armed Bandit, which determines the most appropriate behavioral nudge to display at each recommendation event. Each bandit chooses the best nudge with respect to its known reward while still exploring alternative nudges in order to achieve optimal behavior, consistent with adaptive learning and online decision optimization models [41]. The chosen nudge mechanism, then, constitutes the last output of the system based on (i) personalized signals from the long-term profile of the user and (ii) recent feedback from the behavioral signals [42]. A hybrid design was chosen, combining static personalization (through the TWN) with adaptive behavioral learning (through the MAB). Fig. 3 shows the overall architecture and data flow, illustrating the embedding generation, bandit-based decision making, and nudge deployment.

We also present a formalized workflow of the hybrid model to improve the clarity and reproducibility of its proposed architecture. Using the Two-Tower Network to generate user and product embeddings, relevance scores, and

Algorithm 1: Two-Tower Network Training Pipeline.

Input: User dataset U, Product dataset P, Interaction dataset I
Output: User embedding function $f(u)$, Product embedding function $g(p)$, Relevance score $s(u,p)$

1. Load U, P and I, and build unified interaction dataset D
2. Dataset preprocessing $\rightarrow$ Missing values + Categorical attribute encoding
3. Perform PCA on User & Product Numerical Features to Get Compact Embeddings.
4. Combine PCA features with D to Construct model inputs (X_{user} , X_{product} , y)
5. Address data imbalance and quality:
 - a. Impute missing values
 - b. Apply oversampling (e.g., SMOTE)
 - c. Calculate class weights for training
6. Split datasets into training and testing partitions
7. Define Two-Tower architecture:
 - a. User tower $\rightarrow$ user embedding $f(u)$
 - b. Product tower $\rightarrow$ product embedding $g(p)$
 - c. Concatenate embeddings and compute relevance score $s(u, p)$
8. Train model using weighted loss and test set evaluation.
9. Generate predicted relevance scores for downstream integration with bandit policy

End Algorithm

adaptive nudge selection by incorporating relevance scores in the Multi-Armed Bandit. This stepwise demonstration describes the basic functionality of the system and explains the process of integrating static personalization and dynamic learning at the time of deployment.

The static personalization engine is TWN output, where predicted relevance scores for each user-product pair are generated from TWN. These scores are then provided to the multi-armed bandit as prior utilities for exploration-exploitation. Table 1 provides a summary of the operational parameters for training and deploying TWN.

The selected architecture (128–64–128) is aligned with standard dual-encoder architecture in recommender systems. The period of 20 epochs for training was chosen in an empirical manner from the graphs of validation loss, which suggested that training settled quickly. Dimensionality reduction through PCA was set at a level that maintained $> 95\%$ total variance. Class imbalance led to the application of SMOTE oversampling in parallel to balanced class weights. In this study, the Two-Tower Network has been built and developed from scratch and trained from scratch. The user and item towers were initialized with no pretrained weights and were trained offline for each user by passing the entire behavioral and profile dataset recorded during the study period. These relevance scores constitute the static personalization

TABLE 1. Two-Tower Network Configuration Parameters

COMPONENT	SETTING	NOTES
User tower hidden layer	128 units, relu	Standard dual encoder setting
User embedding layer	64 units, relu	Product tower hidden layer
Product tower hidden layer	128 units, relu	Symmetric with user tower
Product embedding layer	64 units, relu	Latent product representation
Output layer	Sigmoid	Binary relevance prediction
Optimizer	Adam (default lr=0.001)	No tuning performed
Loss function	Binary cross-entropy	Standard for click/purchase modeling
Batch size	32	Default setting
Epochs	20	Sufficient for convergence
Pca user features	Retain $\geq 95\%$ variance	Automatic component selection
Pca product features	Retain $\geq 95\%$ variance	Automatic component selection
Oversampling	Smote	Address class imbalance
Class weights	Balanced (sklearn)	Improve minority-class learning

layer used to seed the recommendation process. TWN was stationary during deployment, while the Multi-Armed Bandit adapted online based upon user feedback.

F. ACTION SPACE AND REWARD FORMULATION

Our system considers each financial product as an independent arm in a Multi-Armed Bandit (MAB). The Two-Tower Network (TWN) generates an initial relevance score for all user-product pairs, which constitutes the prior estimates for the bandit. The bandit produces a ranked list of products for each recommendation round following an ϵ -greedy policy. For illustrative simulation purposes, the reward signal used to update the bandit policy is modeled as stochastic feedback sampled from a Gaussian distribution centered around the expected utility of the selected product.

The value estimate $Q(a)$ for each product a is updated using an incremental stochastic rule:

$$Q_{t+1}(a) = Q_t(a) + \frac{1}{N_t(a)} (R_t - Q_t(a))$$

$$a_t = \begin{cases} \arg \max_a Q_t(a), & \text{with probability } 1 - \epsilon \\ \text{random arm}, & \text{with probability } \epsilon \end{cases}$$

$$\text{Regret}_t = \mu^* - \mu_{a_t}$$

While the Gaussian reward formulation supports stable simulation and regret analysis, the effectiveness of the proposed system is evaluated using real interaction outcomes, including recommendation-following behavior and purchase events, as reported in the experimental results. Through this

Algorithm 2: Adaptive Nudge Selection via Multi-Armed Bandit (Illustrative Simulation).

The reward signal in this algorithm does not represent actual user feedback but is an illustrative (simulation-level) signal necessary for driving policy updates and showing the interplay of exploration and exploitation.

Input:

$S = \{s(u,p)\}$: TWN relevance scores for user u and product p

P : set of available products

Output:

A_t : ranked recommendation list for interaction round t

1. For each user u do
 - Initialize policy values $\pi[u][p] \leftarrow s(u,p)$
 - Initialize reward estimates $R[u][p] \leftarrow 0$
 - Initialize counters $C[u][p] \leftarrow 1$
 - End for
2. For each interaction round t with user u do
 - Determine exploration rate ϵ
 - Rank products by current policy values $\pi[u][p]$
 - Construct recommendation list A_t using ϵ -greedy logic
 - Present recommendation list A_t to the user and observe selected products B_t
3. For each product $p \in B_t$ do
 - Sample reward r_t from Gaussian distribution
 - Update cumulative reward: $R[u][p] \leftarrow R[u][p] + r_t$
 - Update policy estimate: $\pi[u][p] \leftarrow \pi[u][p] + (1 / C[u][p]) \cdot (r_t - \pi[u][p])$
 - Increment counter: $C[u][p] \leftarrow C[u][p] + 1$
 - End for
4. Compute regret
 - Log interaction
 - Generate next recommendation list A_{t+1} using updated policy values $\pi[u][p]$
- End for
- End Algorithm

combination of TWN-based personalization, Gaussian feedback modeling, incremental learning, and regret evaluation, the MAB adaptively prioritizes the products most likely to align with the user's preferences over time.

G. DEPLOYMENT PIPELINE

The integrated nudge recommendation system was deployed in a split A/B setting through two parallel application environments: App v1 (Control) and App v2 (Treatment). To maintain equivalence in our experiment [43] both applications followed the same simulation flow, product catalog, decision points, and user interface. The only difference was the recommendation engine. App v1 relied on a static rule-based mechanism, whereas App v2 incorporated the Two-Tower

Network for personalized product scoring and the Multi-Armed Bandit for adaptive nudge selection [44]. This design ensured that differences were attributable to the AI-driven intervention [22].

During runtime, App v2 ran the entire inference pipeline: user embeddings were produced through the TWN, relevance scores were calculated, and the bandit chose one among nine nudge mechanisms to accompany the suggested product. We showed the user a selected nudge and the product on an immediate basis, and the system collected the behavioral response after each interaction. In App v1, we maintained the same interaction structure, but nudges were assigned deterministically according to business rules [45]. All aspects of user interaction (recommended products, nudges displayed, selection by the user, appointment times, and purchase outcomes) were recorded via a common backend with Supabase. This level of behavioral log granularity enables the reconstruction of decision trajectories and model testing in digital experimentation [46]. We leveraged these event logs to analyze model performance, calculate reward and regret trajectories, and validate the behavioral influence of the treatment condition [47].

IV. RESULTS AND DISCUSSIONS

Due to the quasi-experimental sequential nature of the study, we interpret results as associations consistent with a treatment effect rather than causal estimates. To control for possible cohort and time-window confounds and better capture treatment-related differences, we present effect sizes with confidence limits and, in addition to descriptive comparisons, regression-adjusted estimates.

A. PARTICIPANT AND EXPERIMENT OVERVIEW

In this study, a true A/B experiment with real human users and a realistic digital environment is described. A total of 276 individuals entered the research funnel by completing the consent process, of whom 254 completed the behavioral questionnaire and 253 completed the simulation journey.

In the first round of testing (26–29 November 2024), 214 participants validated as real bank customers were randomly allocated to Group A (Control) and interacted with App Version 1, a static rule-based system providing non-personalized nudges [7], [48]. The control phase collected baseline behavioral data. The 54-item instrument included seven categories (psychographics, communication tendencies, demographics, financial behavioral patterns, preference tendencies, nudge preferences, and timing/frequency sensitivities), representing multidimensional behavioral modeling approaches [6]. Simulated interactions and transaction logs for browsing a product and responding to a nudge were captured following standard logging techniques [49], [50].

The resulting dataset from Group A served as the training foundation for the AI-driven Nudge Recommendation Model. The second test (2 April – 19 May 2025) of 174 Group B

(Treatment) verified bank customers interacting with the AI-enabled system was conducted after the model was placed in production in App Version 2 [51]. The two phases used different groups to avoid carryover effects and preserve internal validity.

B. BASELINE PERFORMANCE AND BEHAVIORAL PATTERNS IN GROUP A (CONTROL)

To better understand the sources of performance improvements, we extend the comparison beyond the rule-based baseline by including a TWN-only ablation. This enables a three-way comparison between the rule-based system, static personalization via TWN, and the proposed TWN+MAB+nudge system. Group A interacted with App Version 1, a rule-based prototype without personalized nudges or AI-driven recommendation logic. This group included 214 verified bank customers whose behavioral patterns served as the benchmark. Two data sources were collected: (1) responses to the 54-item behavioral instrument [52]; and (2) logs of behaviors such as product views, responses to static nudges, and mock-purchase outcomes [8].

Of 214 users, 104 made a purchase (48.6%). A total of 29 purchases were in accordance with the system suggestion (13.6%). The app lacked adaptive optimization, and therefore there was no best recommendation mechanism. These results show that the failure of personalization resulted in poor recommendation compliance and limited behavioral influence. Group A data was aggregated to train the Two-Tower Network and Multi-Armed Bandit [53], providing a baseline of user behavior in a conventional static campaign context.

C. PERFORMANCE OF THE AI-DRIVEN MODEL IN GROUP B (TREATMENT)

In the second testing period (2 April – 19 May 2025), the 174 real users interacted with Nudge Simulation App v2, and the AI-driven Nudge Recommendation Model was tested in Group B. In contrast to the fixed-rule context in Group A, app v2 incorporated the hybrid architecture in which the Two-Tower Network (TWN) produced personalized product relevance scores, while the Multi-Armed Bandit (MAB) used streaming reward signals to dynamically adapt nudge selection in real time. This configuration enabled both personalized baseline recommendations and adaptive behavioral nudging throughout the user journey.

The behavioral outcomes demonstrated a clear improvement in decision-making effectiveness. Out of 174 users, 108 completed a purchase (62.07%), higher than the 48.6% baseline observed in Group A. A total of 92 users made purchases that directly matched the nudges generated by the model, yielding a 52.87% recommendation-driven conversion rate. The strongest finding in this study is that 100% (92) of recommendation-driven purchases were aligned with the optimal arm identified by the MAB algorithm. Reduced exploratory behavior and clearer decision paths were also reflected in logs of system interaction. Together, these results constitute substantial evidence that the AI-driven model, with

its integrated TWN personalization and MAB adaptivity, produced nudges that were better matched in context, motivation, and effectiveness. Moving away from static recommendations toward more responsive decision pathways is consistent with previous work showing that adaptive nudges are significantly more effective than static nudges [54].

D. COMPARATIVE ANALYSIS: GROUP A VS GROUP B

From Group A (Control) and Group B (Treatment), the comparative analysis demonstrates a clear and substantial behavioral shift attributable to the implementation of the AI-powered Nudge Recommendation Model. The experiment involved 214 verified bank customers interacting with a rule-based environment (App v1) and 174 verified bank customers engaging with an AI-enabled environment (App v2). The comparison focuses on three key behavioral metrics: (1) total purchase conversion, (2) recommendation-driven conversion, and (3) alignment with best-arm predictions. Within the constraints of a simulated deployment environment and a controlled experimental design, these metrics provide a focused basis for evaluating relative behavioral differences between conditions.

The strongest causal evidence emerges from recommendation-driven behavior. In Group A, only 29 purchases aligned with the system's static rules (13.6%). In contrast, Group B produced 92 purchases aligned with AI-generated recommendations (52.87%), representing nearly a four-fold increase. This result mirrors prior evidence that adaptive nudges and contextually framed recommendations significantly increase compliance with suggested actions [36], [37]. Since all elements of the experiment were held constant except the nudge-generation mechanism, this disparity directly reflects the causal effect of the AI-driven intervention. This causal interpretation should be understood within the bounds of the sequential A/B design and simulated context, which may not fully capture real-world temporal or economic effects.

The Multi-Armed Bandit (MAB) further amplified this effect by adaptively optimizing nudge strategies in real time. The bandit algorithm consistently identified the optimal arm with the highest reward expectation, and all 92 recommendation-driven purchases (100%) converged on this optimal arm. This indicates strong user acceptability of the algorithmic recommendations and aligns with literature showing that adaptive bandit systems tend to concentrate user actions on high-performing arms once stable reward estimates are established [55]. While this concentration reflects short-term behavioral alignment with the learned policy, it does not by itself imply long-term preference stabilization. Conversely, Group A displayed diffused, unguided choice patterns, consistent with non-adaptively optimized environments.

Taken together, these findings show that the hybrid model combining Two-Tower personalization with adaptive nudging via MAB consistently outperforms the rule-based baseline across all major behavioral metrics. The treatment group demonstrated higher engagement, greater responsiveness to

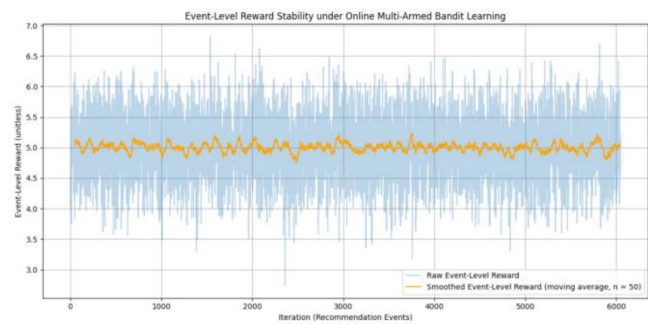


FIGURE 4. Stability of event-level reward trajectories under online multi-armed bandit learning.

recommendations, and substantially stronger behavioral consistency. These results provide strong empirical evidence that AI-driven nudging produces meaningful, consistent, and causally attributable benefits in digital banking campaign performance, reinforcing insights from recent work on personalized decision support systems [54]. Accordingly, these benefits should be interpreted as demonstrating relative effectiveness under the studied experimental conditions rather than unconditional generalizability to real-world financial deployments.

A chi-square test was applied to evaluate whether the difference in recommendation-aligned behavior between groups was statistically significant. The result ($\chi^2 = 6.49$, $p < 0.0108$) confirms that the treatment group’s behavioral shift is unlikely to be driven by random variation, reinforcing the causal effect of the AI-driven nudging mechanism.

E. MULTI-ARMED BANDIT PERFORMANCE: REWARD AND REGRET ANALYSIS

To allow insights into the internal working of MAB in the treatment environment, we provide reward and regret trajectories over time for all recommendation events. Both the raw reward signal and a 50-iteration running average are given in Fig. 4. This figure is not meant to depict a monotonic increase in performance but rather to highlight that the learned policy remains stable under highly noisy, event-level online feedback. Although realizations of individual rewards are highly variable, the time-averaged reward curve remains remarkably stable around a common mean over iterations. This is aligned with the anticipated online bandit learning behavior, which originates from early stochasticity due to exploration, and then stable reward levels can be maintained in later rounds given enough observational evidence. Fig. 4 is a validating test of operability, and such decay in all states could be captured when there are continuous online updates, even though MAB would not deviate as illustrated because improvements for performance achieved by it and policy’s effectiveness are more directly determined through regret analysis (as depicted in Fig. 5) and outcome-based metrics revealed in Table 2.

More information on the quality of the model decision-making can be found in the regret analysis in Fig. 5. Specifically, in this experiment, regret was calculated on a

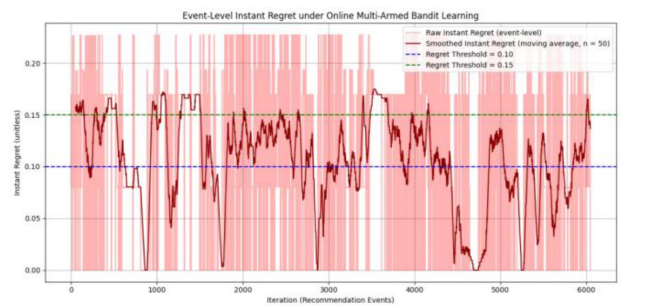


FIGURE 5. Smoothed instant regret over round (Threshold-based evaluation).

TABLE 2. A/B Testing Metrics for Control and Treatment Groups

METRIC	GROUP A	GROUP B	
		TWN	TWN+MAB+ NUDGE SYSTEM
Users in funnel (consent)	214	174	174
Purchases	104	106	108
Purchase Rate	104 / 214 = 48.6% (95% CI: 41.9–55.3%)	106 / 174 = 60.91%	108 / 174 = 62.07% (95% CI: 54.9% – 69.3%)
Purchases from recommendation	29	70	92
Conversion from Recommendation	29 / 214 = 13.6% (95% CI: 9.0–18.2%)	70/174 = 40.23%	92 / 174 = 52.87% (95% CI: 45.5–60.3%)
Purchases from Best Arm	N/A (no best arm logic)	N/A (no best arm logic)	92 / 92 = 100%

per-recommendation-event basis as the difference between the reward obtained for the nudge selected by the bandit and the reward that would have been obtained had the empirically best-performing nudge been opted for instead. This results in an oscillating curve across the interaction window, because this metric represents the quality of an instantaneous decision rather than a long-horizon cumulative formulation. The smoothed regret trajectory stays well under our threshold of 0.10 and 0.15, demonstrating that the bandit rarely strayed far from the empirically optimal nudge in recommendation. The occasional spikes correspond to intentional exploratory behavior, which is a signature and normal feature of adaptive learning systems that are designed to avoid premature convergence [56].

F. BEHAVIORAL INTERPRETATION AND DISCUSSION

The substantial increase in recommendation-aligned purchases (13.6% in Group A versus 52.87% in Group B) indicates that the AI-driven recommendation model significantly influenced users’ decision processes by directing them toward system-optimized options. The experimental design supports a causal interpretation within the setting studied, as the recommendation engine was the only manipulated

variable, users were independently assigned without cross-exposure, and recommendation-following behavior increased by more than fourfold in the treatment group.

For additional context on the size of the behavior effect observed, we provide an effect-size estimate for following recommendation behavior. The prescription adherence probability of the control group increased from 13.6% to 52.87% in the treatment group. This value corresponds to an odds ratio of approximately 7.15, implying that users who were subjected to the AI-based nudge system will be over seven times more likely to mimic a rule-based environment. The magnitude of this effect is sufficiently large that it decreases the possibility that the observed behavioral change derives entirely from time- or cohort-based confounds and is consistent with the framing of our study as an ongoing quasi-experiment.

Taken together, these factors suggest that the observed behavioral differences are primarily attributable to the AI-based nudging mechanism rather than to user composition or experimental artifacts. This interpretation, however, remains bound by the sequential A/B design and simulated deployment context. Within these constraints, the results indicate that combining static personalization through a Two-Tower Network with adaptive optimization via a Multi-Armed Bandit enhances user responsiveness to personalized recommendations by jointly addressing individual relevance and momentary decision context.

While the proposed AI-driven nudge system demonstrates measurable behavioral effects under controlled experimental conditions, real-world deployment would require explicit Responsible AI guardrails. These include risk-tier caps to limit high-impact recommendations, lightweight explanation mechanisms to support user understanding, continuous fairness monitoring across user segments, and opt-out or override options to preserve user autonomy. In addition, periodic auditing of model behavior and outcomes would be necessary to ensure accountability, transparency, and regulatory alignment in financial applications.

V. CONCLUSION

Using a simulated digital banking environment, results offer empirical evidence for the effectiveness of AI-driven nudging in increasing recommendation-following behavior and purchase outcomes. The proposed Nudge Recommendation Model that uses a Two-Tower Network to incorporate static personalization and a Multi-Armed Bandit for dynamic behavioral optimization yielded results whose predictive distributions were drastically different from those of the rule-based baseline. In our A/B experimental control, the treatment had a 62.07% purchase rate and the control group, 48.6%. More importantly, recommendation-following behavior rose from 13.61% in the control to 52.87% under AI, and this increased behavioral alignment was greater than fourfold.

These outcome-level improvements should be interpreted considering the study's experimental design and limitations. While the observed behavioral effects are statistically significant ($\chi^2 = 6.49$, $p < 0.0108$), they are derived from

a simulated context in which real financial risk and long-term behavioral adaptation were not fully present. Therefore, the outcomes really indicate that AI-based nudging can be operationalized and has a relative impact when applied under restricted conditions that deployment will lead to comparable results. Internal learning diagnostics complement these results. Treatment-group recommendation-aligned purchases always agreed with the bandit's best arm, while regret curves revealed nearly constant exploration-exploitation without policy-averse divergence. Such signals affirm the internal learning of the system and complement the outcome-based evaluation.

Collectively, our work provides a reproducible and interpretable approach for combining machine learning models with behavioral economic frameworks in the design of digital personalization systems. Thus, interpretations of our reported findings could be clarified as that of evidence supportive (not confirmatory) of a causal role in the studied context (not all deployment). Future application involves generalization to other, possibly larger and more diverse, user populations; a new rewards definition consistent with domain-specific goals; and further precautions tackling fairness, transparency, and long-term induced behaviors.

LIMITATIONS

Despite contributing important evidence of the benefit of AI-driven nudging, there are limitations to our study. We conducted our experiments in a simulated mobile banking environment and not in the actual financial application, leading to low ecological validity because users were not involved in real monetary losses. The personalization model was limited because it was trained on a medium-sized and domain-specific corpus, making generalizability weaker for users where past behaviors are sparse or demographic profiles are underrepresented in the training data. The A/B testing design equally depended on two non-overlapping deployment windows, which implies that time-based effects cannot be completely discarded, even though the two environments were the same in flow and interface. The study, therefore, should be viewed as a quasi-experiment with temporal or cohort (residual) influences on the observed measures, even if operational control was achieved. Lastly, although the use of AI for nudging has obvious advantages, ethical considerations should not be overlooked; due to its personalized models, the system learns from individual behavioral patterns, this entails challenges for both personal autonomy and fairness. While these risks did not arise in this study, deployment in the real-world would require safeguards preventing the differential disadvantage of certain user groups through increased personalization, or decreased transparency in decision generation over time.

FUTURE WORK

To enhance the impact of the proposed Nudge Recommendation Model, we suggest conducting experimental validation in live mobile banking settings to evaluate how it holds up under actual financial consequences, contextual signals, and

real behavioral trajectories. Putting the system in production would allow assessment of sustained engagement, fairness trajectories, and long-term learning behaviors, including model drift and nudge fatigue. Future work on the adaptive layer can look at contextual or hierarchical bandit models, reinforcement learning methods, or hybrid symbolic–neural systems that incorporate domain rules with behavioral optimization. Finally, as the nudging approach using AI becomes prevalent in financial applications, more work is needed to identify ethical, regulatory, and transparency challenges.

DATA, CODE AVAILABILITY, AND AUTHOR CONTRIBUTIONS

The processed datasets underlying this study are publicly available from Zenodo at <https://zenodo.org/records/16088582>, consisting of anonymized and aggregated experimental outputs used for analysis and figure generation without exposing raw or confidential user data. The source code for the backend services, recommendation logic, and experimental scripts is accessible via the project's Git repository at <https://github.com/nudge-deploy/NudgeCronjob>, enabling reproduction of the reported figures and tables; direct access to the production Supabase infrastructure remains restricted due to security and regulatory considerations. This work was conducted as part of the doctoral dissertation of the lead author, Idha Kristiana, who led the conceptualization, methodology design, data analysis, model development, and manuscript preparation. Harjanto Prabowo, Ford Lumban Gaol, and Nunung Nurul Qomariah served as academic supervisors, providing intellectual guidance, validation, and critical review throughout the research process. All authors reviewed and approved the final manuscript.

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